



Prabhat Kumar

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EDUCATION

Indian Institute of Technology, Kharagpur

Bachelor of Science in Mathematics and Computing

CGPA: 7.29

EXPERIENCE

Software Engineering Intern

May 2026 – July 2026

Paycraft

github.com/Prabhat-190/pbf-generator

- Built a **Java 21** Spring Boot service that generates ABT-to-NLP settlement files in Header, Record, and Trailer format. Validated transit transactions and tracked every file through a **7-stage** path from creation to callback completion.
- Designed **5 REST APIs** for generation, status lookup, callbacks, and health checks, then stored results with Spring Data JPA. Persisted file metadata, per-record outcomes, and **4-digit** daily sequences in an H2 database.
- Implemented trailer checks, **SHA-256** checksums, ZIP packaging, mock SFTP delivery, and year-month archival. Added SLF4J request-id logging, OpenAPI docs, **JUnit 5** tests, and Docker packaging for Railway deployment.

PROJECTS

Bachelor's Thesis Project: Machine Learning-Based Stock Prediction and Portfolio Optimization

Ongoing

Python, Pandas, NumPy, Scikit-learn, Keras, Matplotlib

- Building a machine learning pipeline that forecasts next-day **Nifty 50** returns and then forms an optimized stock portfolio. Feeding historical OHLCV prices and **5** technical indicators into the forecast stage before capital is allocated.
- Training **3** models, Linear Regression, Random Forest, and LSTM, with walk-forward validation to block look-ahead leakage. Ranking mean-variance portfolios by **Sharpe ratio** and maximum drawdown against equal-weight and index baselines.

Algo Trade Simulator

Live Project

Python, Dash, Plotly, Redis, WebSockets, Docker, Unicorn, Nginx

- Developed a live trading simulator that reads **Level-2** order-book streams and estimates execution cost for market orders. Modeled **6** cost drivers: slippage, market impact, exchange fees, liquidity, volatility, and maker-taker mix.
- Designed a **4-layer** Redis publish-subscribe pipeline for ingestion, book updates, cost analysis, and live charts. Deployed with Unicorn and Nginx, added Redis failover, and wrote more than **100** tests for feeds and persistence.

Amazon FinAI: Smart Checkout and Conversational Assistant

GitHub

React, Node.js, FastAPI, MongoDB, Redis, ChromaDB, OpenAI, Docker

- Developed a checkout platform that routes payments and answers user questions with a retrieval-augmented assistant. Combined wallet, budget, and policy context so the assistant can return grounded financial answers.
- Ranked payment methods on **3 Redis** signals: gateway success, cashback value, and gift-card expiry. Extracted discounts from bank-offer images with GPT-4 Vision and shipped **3** services with Docker Compose.

Fraud Shield AI

Live Project

Python, Scikit-learn, Pandas, NumPy, Streamlit, CCXT

- Built a fraud detection model that scores each transaction from **0 to 100 percent** risk using a Random Forest classifier. Tuned the decision threshold on the precision-recall curve so genuine payments are not blocked by false alerts.
- Engineered **5** features: amount, time, cardholder age, location, and merchant category for each scored transaction. Deployed a Streamlit dashboard for live scoring and used CCXT market data with an automatic offline fallback.

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, SQL

Frontend: React.js, Next.js, HTML, CSS

Backend: Spring Boot, Spring Data JPA, REST APIs, Node.js, Express.js, FastAPI, WebSocket, Microservices

Machine Learning: Scikit-learn, Keras, Pandas, NumPy, LSTM, Time-Series Forecasting, Portfolio Optimization

Databases and Tools: MySQL, MongoDB, Redis, H2, ChromaDB, Docker, Git, Gradle, JUnit, SLF4J, Unicorn, Nginx, OpenAPI, Postman

Computer Science: Object-Oriented Programming, Operating Systems, Computer Networks, System Design

RELEVANT COURSEWORK

Computer Science: Algorithms, Data Structures, DBMS, File Organization, Computer Organization, Theory of Computation, Big Data Analysis.

Mathematics: Optimization, Probability and Statistics, Stochastic Processes, Linear Algebra, Discrete Mathematics, Graph Theory.

ACHIEVEMENTS

- Rated Codeforces Expert with a maximum rating of **1736** and Global Rank **245** among **25,673** participants in Round 1093.
- Solved more than **500** data structures and algorithms problems on LeetCode, Codeforces, CodeChef, and HackerRank.
- Awarded the Reliance Foundation Undergraduate Scholarship, selected among the top **0.8 percent** of over **120,000** applicants.
- Secured a Top **8 percent** JEE Advanced rank, a Top **1.2 percent** JEE Main rank, and the HackerRank Intern certification.